MODULE 2

DATA EXTRACTION

Feature Selection • Filters • Wrappers • Entropy • Decision Trees • Random Forest • Dimensionality Reduction • SVD • PCA

Study Notes

This module introduces techniques used to select useful features, reduce unnecessary dimensions, and build tree-based machine learning models. The notes are written in an exam-friendly format with definitions, working steps, advantages, limitations, and examples.

1. Feature Selection

Feature selection is the process of selecting the most relevant input variables (features) from a dataset while removing irrelevant or redundant features.

Why feature selection is needed

- Reduces the number of input variables.
- Removes irrelevant or noisy features.
- Can improve model accuracy and generalization.
- Reduces training time and memory usage.
- Makes the model easier to understand and interpret.

Example: For predicting student performance, features such as attendance, previous marks, and study hours may be useful, while an unrelated ID number usually provides no useful information.

Main approaches: Filter methods, Wrapper methods, and Embedded methods.

2. User Retention

User retention refers to the ability of a product, application, or service to keep users active over a period of time. In data analysis, retention is often studied using historical user activity data.

Retention rate

A simple retention-rate calculation is:

> **Retention Rate = (Number of users retained during a period / Number of users at the beginning of the period) × 100**

- Example: If 800 users start a month and 600 are still active at the end, retention rate = 75%.

Features useful for retention analysis: login frequency, session duration, purchases, subscription status, last activity date, and number of interactions.

Relation to feature selection: selecting the strongest behavioral features can help a model identify users who are likely to remain active or leave.

3. Filter Methods

Filter methods select features using statistical properties of the data before a machine-learning model is trained. They are generally fast because they are model-independent.

Common filter techniques

- Correlation – identifies strongly related variables.
- Chi-square test – commonly used for categorical features and categorical targets.
- ANOVA / F-test – can compare differences between groups.
- Mutual information – measures how much information one variable provides about another.
- Variance threshold – removes features with very low variance.

Advantages: fast, simple, scalable, and independent of the final model.

Limitation: a filter method may select features that look individually useful but are not optimal when combined with other features.

4. Wrapper Methods

Wrapper methods evaluate different subsets of features by actually training and evaluating a machine-learning model. The model's performance is used to decide which feature subset is better.

Common wrapper techniques

- Forward Selection – starts with no features and adds useful features one by one.
- Backward Elimination – starts with all features and removes less useful features.
- Recursive Feature Elimination (RFE) – repeatedly removes the least important features.

Basic process: Choose a feature subset → train the model → evaluate performance → change the subset → repeat → select the best subset.

Advantages: considers feature interactions and is closely related to the chosen model.

Limitation: computationally expensive because many models may need to be trained.

5. Entropy

Entropy measures the impurity or uncertainty in a dataset. It is commonly used in decision trees to decide which feature should be used for splitting.

For a classification problem, entropy is:

$$\textbf{Entropy}(S) = -\sum p_i \log_2(p_i)$$

- p_i = proportion of samples belonging to class i.
- Entropy = 0 when all samples belong to one class.
- Higher entropy means greater uncertainty or impurity.

Example: If a node contains only Positive examples, its entropy is 0. If Positive and Negative examples are equally divided, entropy is 1.

6. Decision Tree Algorithm

A decision tree is a supervised learning algorithm that makes decisions by repeatedly splitting data according to feature values. It can be used for classification and regression.

Important parts

- Root node – first and topmost decision.
- Internal node – represents a decision or test on a feature.
- Branch – represents an outcome of a decision.
- Leaf node – represents the final prediction.

Working steps

- Start with the complete training dataset.
- Calculate a splitting criterion such as information gain or Gini impurity.
- Choose the best feature/split.
- Split the data into child nodes.
- Repeat the process for the child nodes.
- Stop when a stopping condition is reached, such as maximum depth or sufficient purity.

Information Gain:

Information Gain = Entropy(parent) − Weighted Entropy(children)

Advantages: easy to interpret, handles nonlinear relationships, and generally requires little feature scaling.

Limitations: can overfit, especially when the tree becomes very deep; pruning or depth restrictions can help.

7. Random Forest

Random Forest is an ensemble learning algorithm that combines many decision trees to produce a more robust prediction.

Working principle

- Create multiple training samples using bootstrap sampling.
- Build a decision tree for each sample.
- At each split, consider a random subset of features.
- For classification, combine tree predictions using majority voting.
- For regression, commonly average the predictions.

Why it works: individual trees can have high variance. Combining many diverse trees generally improves stability and reduces overfitting compared with a single unrestricted tree.

Advantages: strong performance, handles many features, less sensitive to noise than a single tree, and can estimate feature importance.

Limitation: less interpretable than one decision tree and can require more computation and memory.

8. Dimensionality Problems

Dimensionality refers to the number of features or variables in a dataset. A dataset with many features is called high-dimensional.

Problems caused by high dimensionality

- Curse of dimensionality – data become sparse as the number of dimensions increases.

- Higher computational and storage cost.
- More irrelevant and redundant information.
- Greater risk of overfitting.
- Difficulty in visualizing and interpreting the data.

Solution: dimensionality can be reduced through feature selection or feature extraction techniques such as PCA and SVD.

9. Singular Value Decomposition (SVD)

Singular Value Decomposition decomposes a matrix into three matrices. It is widely used for dimensionality reduction, noise reduction, data compression, and recommender systems.

For a matrix A, SVD is represented as:

$$\mathbf{A = U\Sigma V^T}$$

- U – left singular vectors.
- Σ (Sigma) – diagonal matrix containing singular values.
- V^T – transpose of the right singular-vector matrix.

Basic idea: SVD represents the original matrix using directions that capture different amounts of information. By keeping only the largest singular values, a lower-dimensional approximation can be obtained.

Applications: image compression, text analysis, recommendation systems, noise reduction, and latent-feature extraction.

10. Principal Component Analysis (PCA)

PCA is a dimensionality-reduction technique that transforms correlated features into a smaller set of new variables called principal components.

Key idea

- The first principal component captures the maximum possible variance.
- The second principal component captures the maximum remaining variance while being orthogonal to the first.
- Further components capture progressively smaller amounts of variance.

Typical PCA steps

- Collect the numerical feature matrix.
- Standardize the features when their scales differ substantially.
- Calculate the covariance/correlation structure.
- Find principal directions using eigenvectors (or an SVD-based implementation).
- Rank components by explained variance.
- Keep the desired number of components and transform the data.

Advantages: reduces dimensions, removes some redundancy, supports visualization, and can speed up later modeling.

Limitations: transformed components can be difficult to interpret, scaling matters, and PCA is mainly designed to capture variance rather than directly optimize prediction accuracy.

11. Quick Comparison

The following table helps distinguish the major techniques in this module.

Technique	Main purpose	Model dependent?
Filter	Select features using statistical measures	No
Wrapper	Find feature subset using model performance	Yes
Decision Tree	Prediction using sequential feature-based splits	Yes – it is a model
Random Forest	Combine many decision trees	Yes – it is a model
SVD	Matrix decomposition / dimensionality reduction	No
PCA	Create fewer components that preserve variance	No

12. Important Exam Points

- Feature Selection: selecting relevant features and removing irrelevant or redundant ones.
- Filter: fast, model-independent feature selection based on statistical measures.
- Wrapper: evaluates feature subsets using model performance.
- Entropy: measures uncertainty/impurity in a dataset.
- Decision Tree: uses feature-based splits to reach a prediction at a leaf node.
- Random Forest: combines many decision trees using randomness and aggregation.
- Dimensionality: the number of features in a dataset; high dimensionality can cause sparsity, cost, and overfitting.
- SVD: decomposes a matrix as $A = U\Sigma V^T$.
- PCA: transforms features into principal components ordered by explained variance.